Math 106L In Class 3 Practice

Contents

Improper Integrals 2

Basic Differential Equations 3

Slopefields and Equilibria 4

Separation of Variables 11

Newton's Law of Cooling 12

Mixing 17

Chemical Rates 20

Converting Descriptions to DEs 21

Improper Integrals

For all these, correct notation involving limits is essential. I strongly suggest using the sidebar format we did on the corresponding worksheets. When doing substitutions, be sure to change bounds!

1. Compute $\int_1^\infty \frac{1}{x^3}\,dx$.
2. Compute $\int_1^\infty \frac{1}{\sqrt{x}}\,dx$.
3. Compute $\int_0^\infty e^{-3x}\,dx$.
4. Compute $\int_e^\infty \frac{1}{x(\ln x)^2}\,dx$.
5. Compute $\int_{-\infty}^0 e^{2x}\,dx$.
6. Compute $\int_0^\infty \frac{1}{1+x^2}\,dx$.
7. Compute $\int_1^\infty \frac{\ln x}{x}\,dx$.
8. Compute $\int_0^1 \frac{1}{\sqrt[3]{x}}\,dx$.
9. Compute $\int_0^4 \frac{1}{(x-4)^2}\,dx$.
10. Compute $\int_2^5 \frac{1}{\sqrt{x-2}}\,dx$.
11. Compute $\int_0^1 \ln x\,dx$.
12. Compute $\int_{-1}^1 \frac{1}{x^3}\,dx$.
13. Compute $\int_0^{\pi/2} \sec^2 x\,dx$.
14. Compute $\int_{-\infty}^\infty \frac{x}{(1+x^2)^2}\,dx$.
15. Compute $\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx$.

Basic Differential Equations

Solve the following initial value problems.

1. $\dfrac{dy}{dx} = 3x^2 - 4$, with $y(1) = 2$.
2. $y' = \sin(x) + 2x$, with $y(0) = 3$.
3. $\dfrac{dy}{dx} = e^{3x}$, with $y(0) = 1$.
4. $y' = \sec^2(x)$, with $y\left(\frac{\pi}{4}\right) = 2$.
5. $\dfrac{dy}{dx} = \dfrac{1}{x}$, with $y(e) = 4$ for $x > 0$.
6. $y' = \dfrac{2x}{x^2 + 1}$, with $y(0) = 5$.
7. $\dfrac{dy}{dx} = x\sqrt{x^2 + 9}$, with $y(4) = 0$.
8. $y' = \dfrac{(\ln x)^2}{x}$, with $y(1) = 2$.
9. $\dfrac{dy}{dx} = \ln(x)$, with $y(1) = -1$.
10. $y' = x\cos(x)$, with $y(0) = 1$.
11. $y'' = 12x^2 - 6x$, with $y(0) = 1$ and $y'(0) = 2$.
12. $y'' = e^x$, with $y(0) = 2$ and $y'(0) = 3$.
13. $y'' = -\cos(x)$, with $y(\pi) = 0$ and $y'(\pi) = 1$.
14. $y'' = \dfrac{1}{(x+1)^2}$, with $y(0) = 2$ and $y'(0) = 1$.
15. $y'' = xe^x$, with $y(0) = 1$ and $y'(0) = 0$.

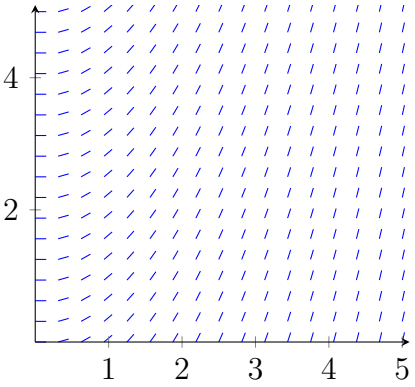
Slopefields and Equilibria

1. Match the following differential equations to their corresponding slope fields. All slope fields are shown in the first quadrant only ($0 \le t \le 5$, $0 \le y \le 5$).

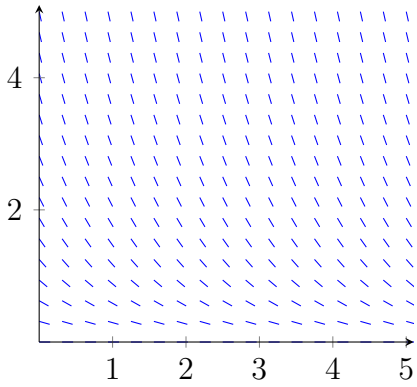
(a) $\dfrac{dy}{dt} = t - y$ (c) $\dfrac{dy}{dt} = -y$ (e) $\dfrac{dy}{dt} = e^{-t}$

(b) $\dfrac{dy}{dt} = \cos(y)$ (d) $\dfrac{dy}{dt} = t + y$ (f) $\dfrac{dy}{dt} = t$

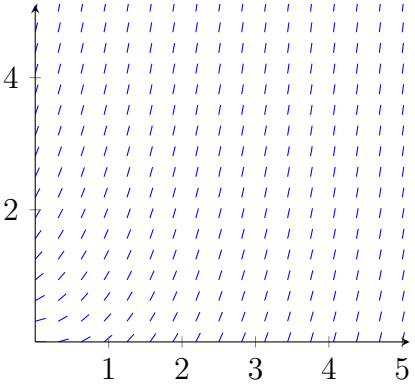
i)



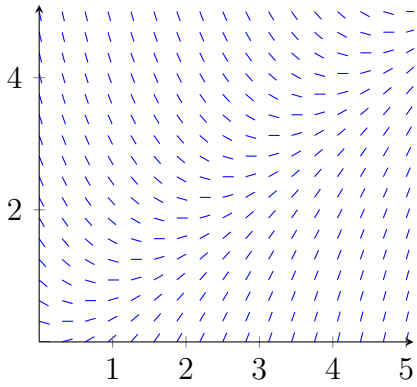
ii)



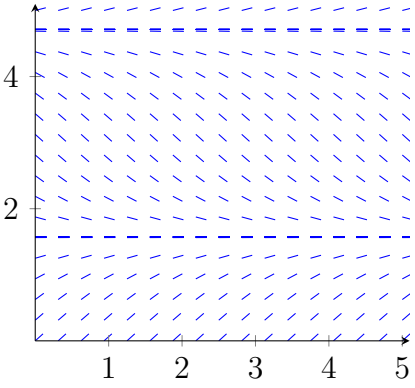
iii)



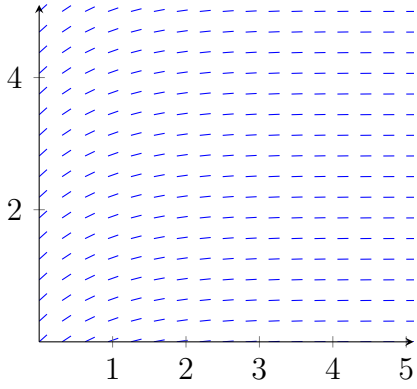
iv)



v)



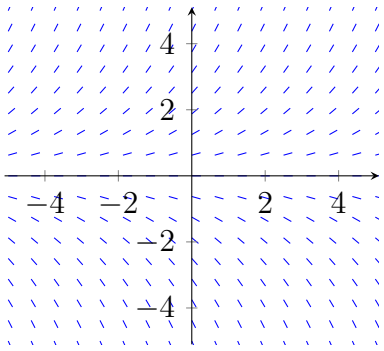
vi)

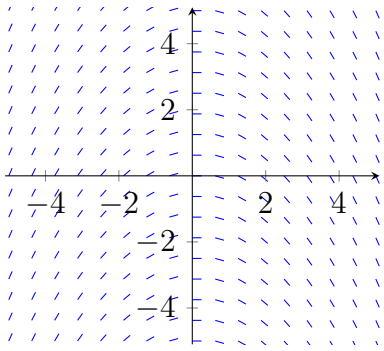


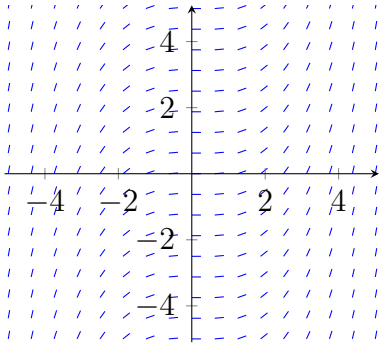
2. Match the following differential equations to their corresponding slope fields. These fields are graphed in all four quadrants ($-5 \le t \le 5$, $-5 \le y \le 5$).

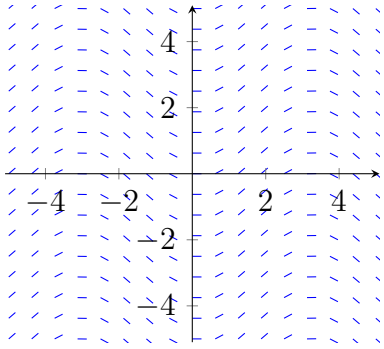
(a) $\dfrac{dy}{dt} = -\dfrac{t}{2}$ (c) $\dfrac{dy}{dt} = \sin(t)$ (e) $\dfrac{dy}{dt} = \dfrac{y}{2}$

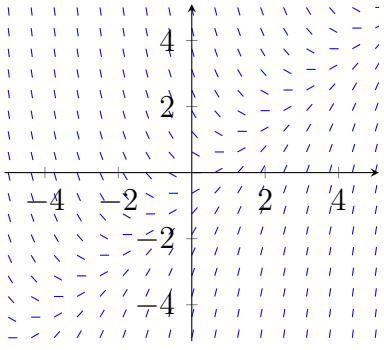
(b) $\dfrac{dy}{dt} = y - t^2$ (d) $\dfrac{dy}{dt} = \dfrac{t^2}{4}$ (f) $\dfrac{dy}{dt} = t - y$

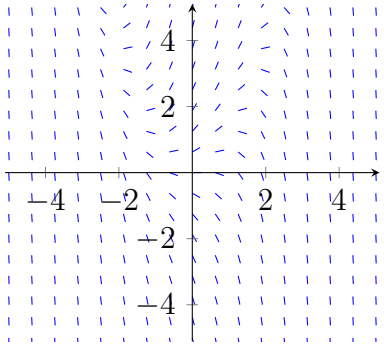
i) 

ii) 

iii) 

iv) 

v) 

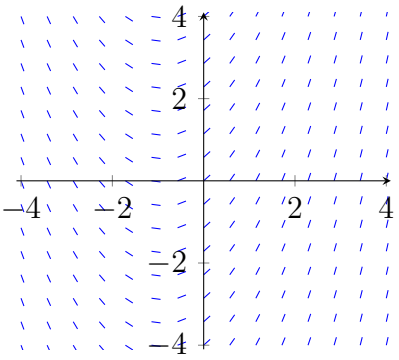
vi) 

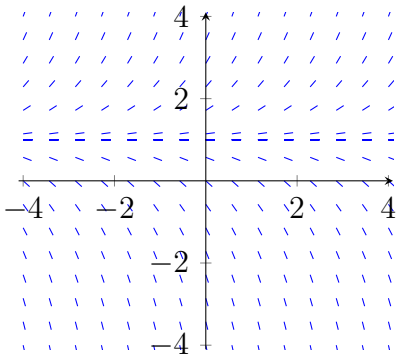
3. Match the differential equations to the slope fields.

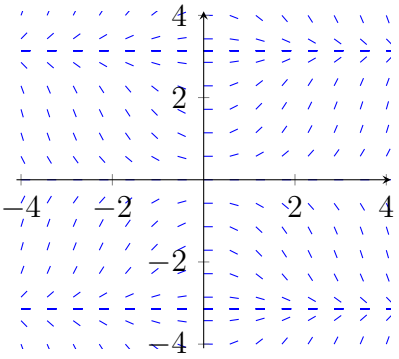
Note: This is not a perfect 1-to-1 matching. Two of the differential equations provided below do not have a corresponding slope field, and two of the slope fields below do not have a corresponding differential equation. If an equation has no match, write "None".

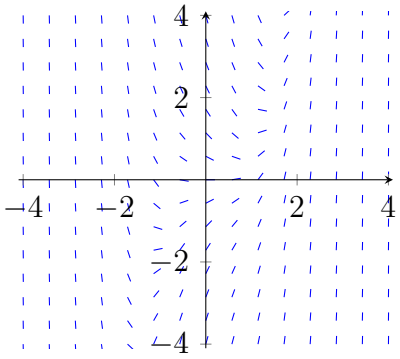
(a) $\frac{dy}{dt} = t^3 - y$ (c) $\frac{dy}{dt} = t^2 y$ (e) $\frac{dy}{dt} = t + 1$

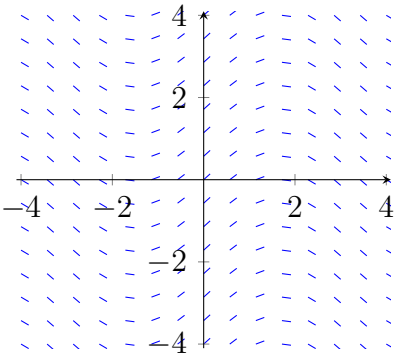
(b) $\frac{dy}{dt} = t\sin(y)$ (d) $\frac{dy}{dt} = y - 1$ (f) $\frac{dy}{dt} = e^t \cos(y)$

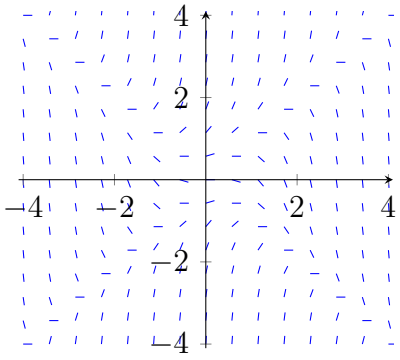
i) 

ii) 

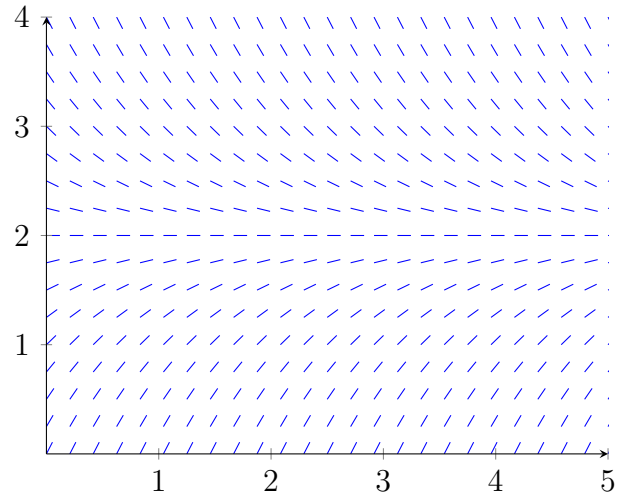
iii) 

iv) 

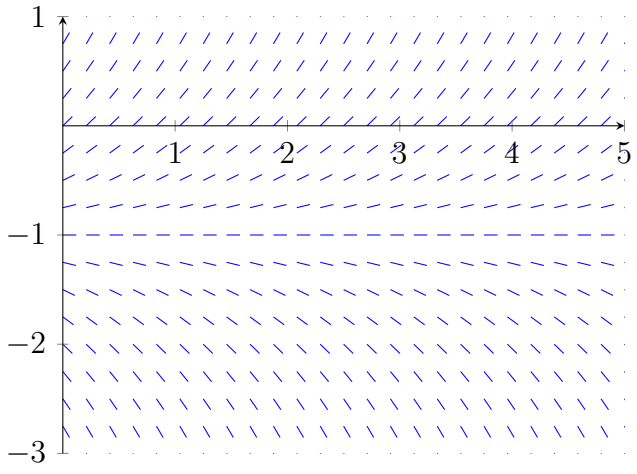
v) 

vi) 

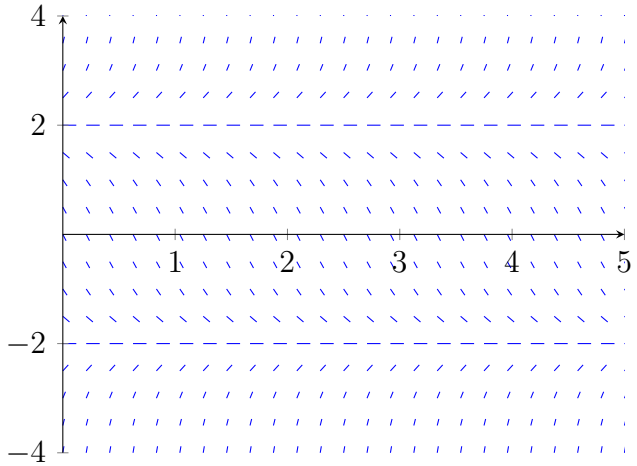
4. Find all equilibria of the differential equation $\frac{dy}{dt} = 2 - y$. For each equilibrium, determine whether it is stable or unstable.



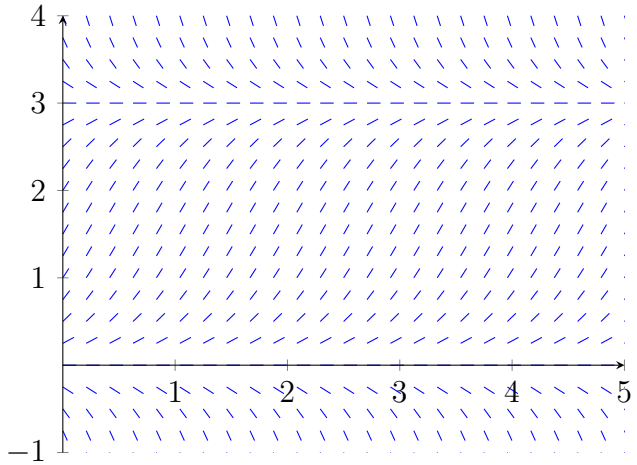
5. Find all equilibria of the differential equation $\frac{dy}{dt} = y + 1$. For each equilibrium, determine whether it is stable or unstable.



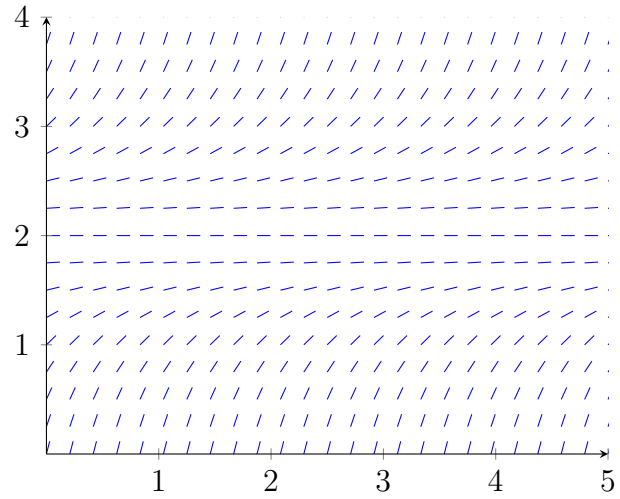
6. Find all equilibria of the differential equation $\frac{dy}{dt} = y^2 - 4$. For each equilibrium, determine whether it is stable or unstable.



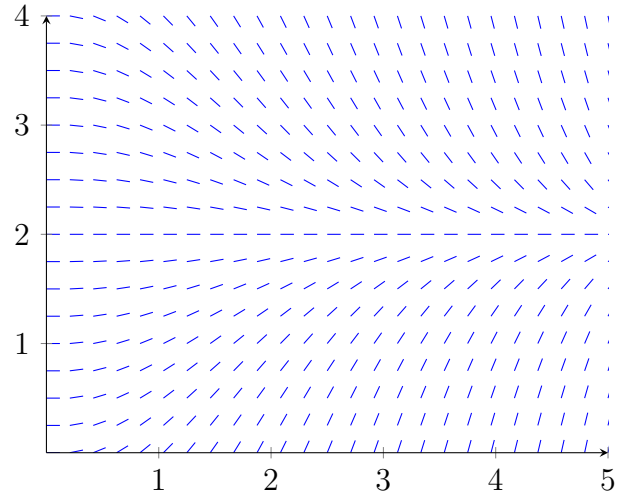
7. Find all equilibria of the differential equation $\frac{dy}{dt} = y(3 - y)$. For each equilibrium, determine whether it is stable or unstable.



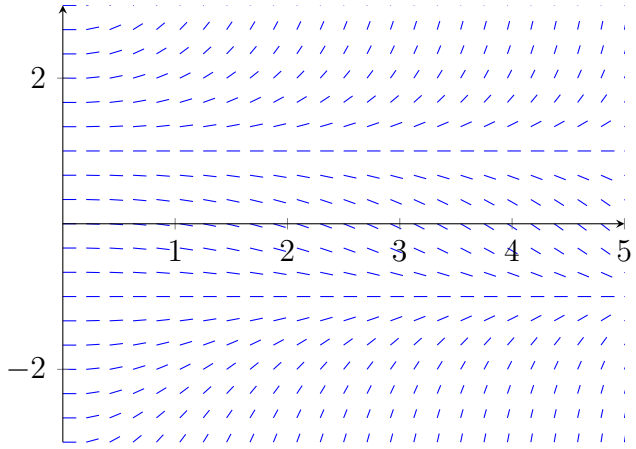
8. Find all equilibria of the differential equation $\frac{dy}{dt} = (y-2)^2$. For each equilibrium, determine whether it is stable or unstable.



9. Find all equilibria of the differential equation $\frac{dy}{dt} = -\frac{t}{2}(y-2)$. For each equilibrium, determine whether it is stable or unstable.



10. Find all equilibria of the differential equation $\frac{dy}{dt} = \frac{t}{4}(y^2 - 1)$. For each equilibrium, determine whether it is stable or unstable.



Separation of Variables

For each of the following initial value problems, determine if the differential equation is separable. If it is, solve the initial value problem. If it is not, say so.

1. $\frac{dy}{dx} = 2xy^2$, with $y(0) = 1$
2. $\frac{dy}{dx} = yxe^{x^2}$, with $y(0) = e$
3. $\frac{dy}{dx} = x + y$, with $y(0) = 1$
4. $\frac{dy}{dx} = \frac{x^2}{y}$, with $y(1) = 2$
5. $\frac{dy}{dx} = \frac{xe^x}{y^2}$, with $y(0) = 3$
6. $\frac{dy}{dx} = \sin(x + y)$, with $y(0) = \pi$
7. $\frac{dy}{dx} = xe^{-y}\ln(x)$, with $y(1) = 0$
8. $\frac{dy}{dx} = \frac{x}{y\ln(y)}$, with $y(0) = e$
9. $\frac{dy}{dx} = \frac{xy + x}{y}$, with $y(1) = 1$
10. $x\frac{dy}{dx} + y = x^2y^2$, with $y(1) = 1$
11. $\frac{dy}{dx} = e^{x-y}$, with $y(0) = \ln(2)$
12. $\frac{dy}{dx} = \frac{2x}{y(x^2+1)}$, with $y(0) = -2$
13. $\frac{dy}{dx} = \frac{1}{x^2ye^{y^2}}$, with $y(1) = \sqrt{\ln(2)}$
14. $\frac{dy}{dx} = \sqrt{xy}$, with $y(1) = 4$
15. $\frac{dy}{dx} = x\cos(x)(y^2 + 1)$, with $y(0) = 1$

Newton's Law of Cooling

1. Suppose that a bowl of hot soup, which has a temperature of 200°F, is placed in a dining room which is at a constant temperature of 70°F. Suppose that we measure time in minutes after the soup is served, and that the constant of proportionality for its cooling is $k = 0.4$ per minute. By writing down and solving an initial value problem, find how long it will take for the temperature of the soup to cool to a consumable 100°F?

2. A cold can of soda at 35°F is taken out of a refrigerator and placed on a patio where the ambient temperature is 85°F. The temperature of the soda increases according to Newton's Law of Cooling with a constant of proportionality $k = 0.5$ when time is measured in hours. By writing down and solving an initial value problem, find how long it will take for the soda to warm up to 60°F.

3. A pizza is removed from a 450°F oven and placed on a counter in a 75°F kitchen to cool. The pizza cools according to a proportionality constant of $k = 0.2$ per minute. Find a formula for the temperature of the pizza as a function of time. How long will it take until the temperature reaches a safe eating temperature of 150°F?

4. A piece of glowing steel at 1200°F is plunged into an oil quench bath whose temperature is kept at a constant 100°F. The temperature of the steel decreases according to Newton's Law of Cooling with a constant of proportionality $k = 1.5$ when time is measured in minutes. Find a formula for the temperature of the steel as a function of time. How long will it take until the temperature reaches 200°F?

5. A carton of ice cream is removed from a 10°F freezer and accidentally left on a table in a 72°F room. The rate of temperature change has a proportionality constant of $k = 0.8$ per hour. By writing down and solving an initial value problem, determine how long it will take for the ice cream to reach 30°F (at which point it begins melting rapidly).

6. A head chef is found dead inside the restaurant's walk-in refrigerator. The refrigerator's temperature is kept at a constant 38°F. The police arrive and measure the temperature of the body at 8:00 AM, finding it to be 68°F. They take a second reading two hours later at 10:00 AM, and the temperature has dropped to 53°F.

   Assume the victim's normal body temperature was 98°F at the exact time of death. Security footage from the hallway shows three employees leaving the walk-in refrigerator that morning:

   - Alice (The Sous Chef) left at 5:00 AM.
   - Bob (The Line Cook) left at 6:00 AM.
   - Charlie (The Manager) left at 7:00 AM.

   By writing and solving a differential equation using Newton's Law of Cooling, determine the exact time of death. Which suspect should the police arrest?

7. The following four graphs show the temperature $T(t)$ of four different objects cooling over time. All four objects start at an initial temperature of 100°C.

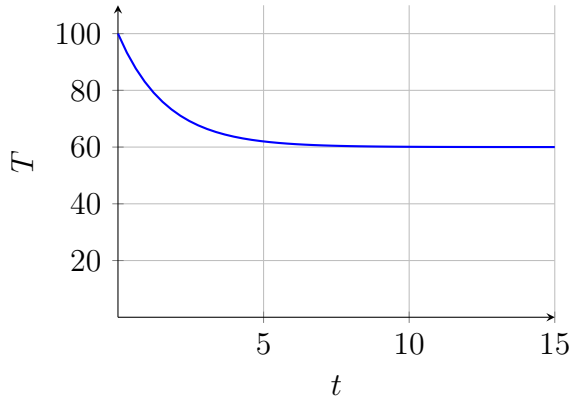
   Match each graph to its corresponding differential equation below by considering the role of the ambient temperature (T^*) and the relative cooling rate (k).

   Note: Two of the differential equations provided do not match any of the graphs. If an equation has no match, write "None".

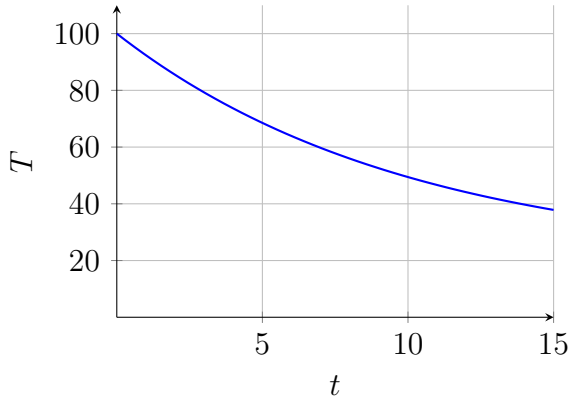
(a) $\frac{dT}{dt} = -0.6(T - 60)$ (c) $\frac{dT}{dt} = -0.1(T - 60)$ (e) $\frac{dT}{dt} = -0.6(T - 80)$

(b) $\frac{dT}{dt} = -0.1(T - 20)$ (d) $\frac{dT}{dt} = -0.6(T - 20)$ (f) $\frac{dT}{dt} = -0.1(T - 0)$

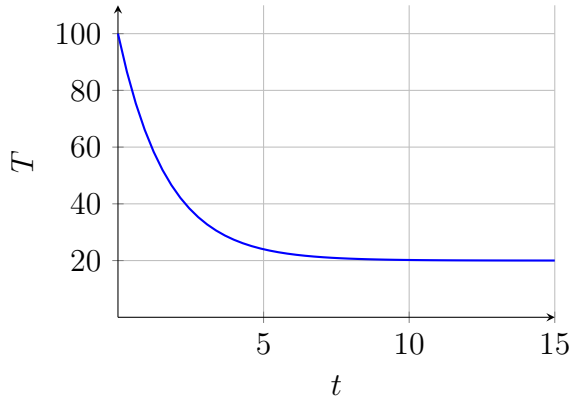
i)



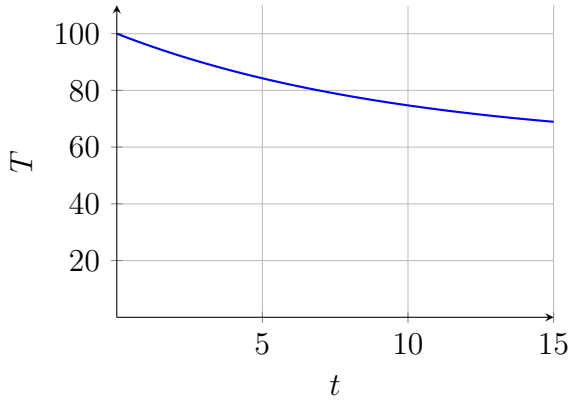
ii)



iii)



iv)



8. Match each physical scenario described below with the differential equation that best models its temperature, $T(t)$.

   When matching, consider the role of the ambient temperature (T^*) and how the material properties (like insulation) affect the relative cooling/heating rate (k).

   Note: Two of the differential equations provided do not match any of the scenarios. If an equation has no match, write "None".

(a) $\frac{dT}{dt} = -0.5(T - 90)$ (c) $\frac{dT}{dt} = -0.5(T - 35)$ (e) $\frac{dT}{dt} = -0.02(T - 68)$

(b) $\frac{dT}{dt} = -0.5(T - 68)$ (d) $\frac{dT}{dt} = -0.02(T - 90)$ (f) $\frac{dT}{dt} = -0.02(T - 180)$

i) A hot mug of coffee (initially 180°F) is poured into a thin, uninsulated paper cup and left on a table in a 68°F living room.

ii) A hot mug of coffee (initially 180°F) is poured into a highly insulated, double-walled vacuum thermos and left on a table in a 68°F living room.

iii) A cold soda (initially 35°F) is left in a thin aluminum can (which conducts heat rapidly) on a sunny patio where the temperature is 90°F.

iv) A cold soda (initially 35°F) is placed inside a thick, heavy-duty foam koozie and left on a sunny patio where the temperature is 90°F.

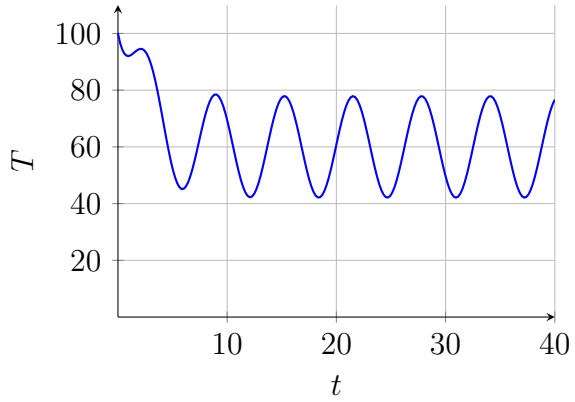
9. The following four graphs show the temperature $T(t)$ of four different objects. Each object starts at an initial temperature of 100°F and is placed in an environment where the ambient temperature fluctuates according to $T^*(t) = 40\sin(t) + 60$.

   Notice that the x-axes have been scaled differently for each graph so that the initial rate of cooling visually appears identical. Match each graph to its corresponding differential equation by observing the **amplitude of the eventual sinusoidal steady-state**.

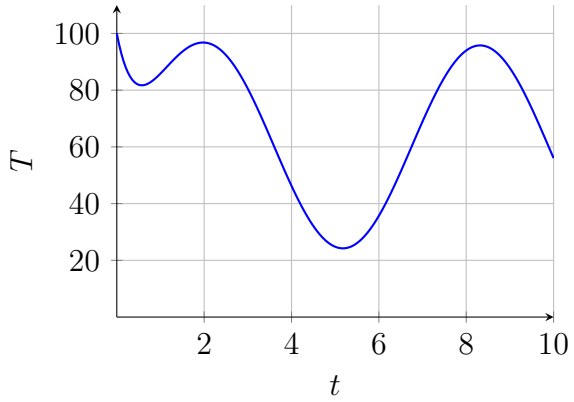
(a) $\dfrac{dT}{dt} = -2\big(T - (40\sin(t) + 60)\big)$ (c) $\dfrac{dT}{dt} = -0.5\big(T - (40\sin(t) + 60)\big)$

(b) $\dfrac{dT}{dt} = -1\big(T - (40\sin(t) + 60)\big)$ (d) $\dfrac{dT}{dt} = -0.25\big(T - (40\sin(t) + 60)\big)$

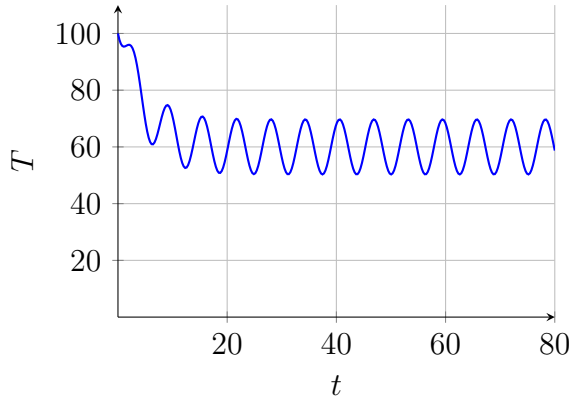
i)



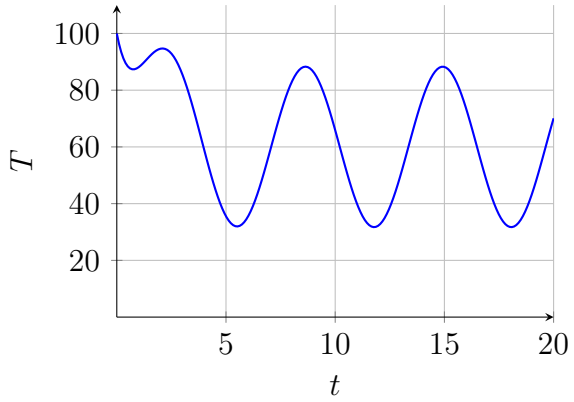
ii)



iii)



iv)



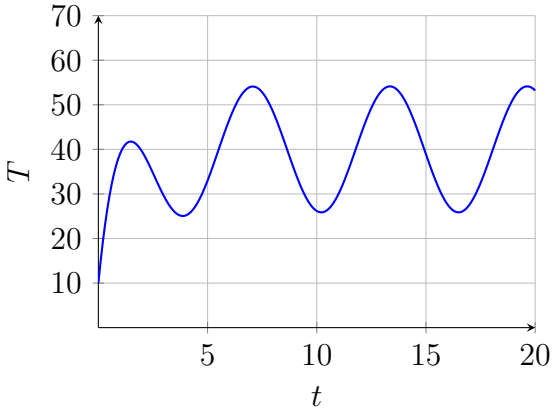
10. The following four graphs show the temperature $T(t)$ of four different objects starting at 10°F, placed into an environment where the temperature fluctuates according to $T^*(t) = 20\cos(t) + 40$.

    Once again, the x-axes have been scaled so the initial warming phase visually matches across all plots. Match the graphs to their differential equations by looking at the steady-state amplitude.

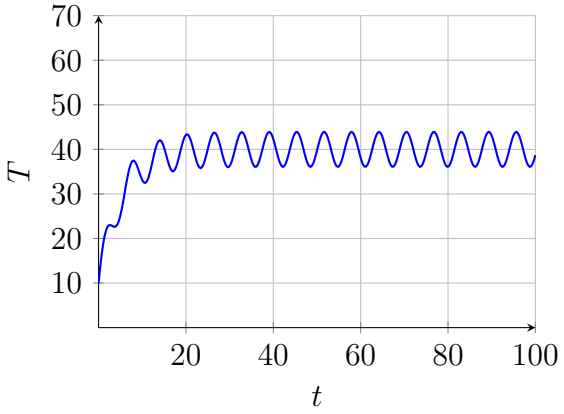
    (a) $\dfrac{dT}{dt} = -2\big(T - (20\cos(t) + 40)\big)$ (c) $\dfrac{dT}{dt} = -0.5\big(T - (20\cos(t) + 40)\big)$

    (b) $\dfrac{dT}{dt} = -1\big(T - (20\cos(t) + 40)\big)$ (d) $\dfrac{dT}{dt} = -0.2\big(T - (20\cos(t) + 40)\big)$

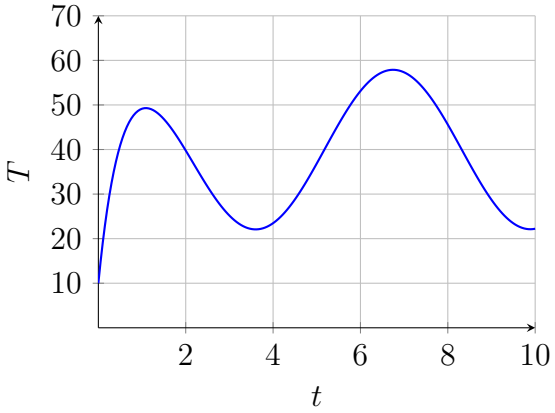
i)



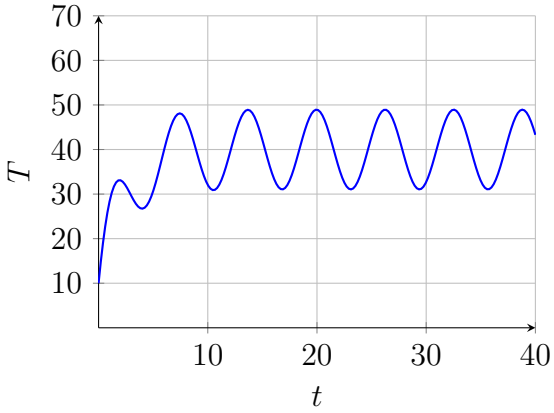
ii)



iii)



iv)



Mixing

1. A 500 gallon tank is initially filled with pure water. Every minute, 5 gallons of brine containing 0.2 pounds of salt per gallon are pumped into the tank. The well-mixed solution is pumped out at the same rate of 5 gallons per minute. Find a differential equation and initial condition satisfied by $S(t)$, the amount of salt in the tank at time t. Then, solve the initial value problem for $S(t)$.

2. A 1000 liter tank initially contains 50 kilograms of dissolved salt. Pure water is pumped into the tank at a rate of 10 liters per minute. The well-mixed solution is pumped out at the exact same rate. Find a differential equation and initial condition satisfied by $S(t)$, the amount of salt in the tank at time t. Then, solve the initial value problem for $S(t)$.

3. A 200 gallon tank initially contains 40 pounds of salt dissolved in water. A brine solution containing 0.5 pounds of salt per gallon is pumped into the tank at a rate of 4 gallons per minute. The well-mixed solution is pumped out at a rate of 4 gallons per minute. Find a differential equation and initial condition satisfied by $S(t)$, the amount of salt in the tank at time t. Then, solve the initial value problem for $S(t)$.

4. A 100 liter tank is initially filled with a solution containing 10 kilograms of a chemical dye. A new solution containing 0.4 kilograms of dye per liter is pumped in at a rate of 2 liters per minute. The well-mixed solution is pumped out at 2 liters per minute. Find a differential equation and initial condition satisfied by $S(t)$, the amount of dye in the tank at time t. Then, solve the initial value problem for $S(t)$.

5. The following four graphs show the amount of salt $S(t)$ (in pounds) in four different tanks over time t (in minutes). All four tanks start with pure water, meaning $S(0) = 0$.

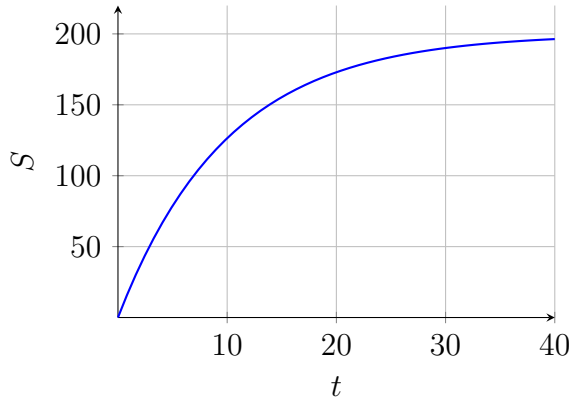
   Match each graph to its corresponding differential equation by considering the **equilibrium value** (which represents the long-term amount of salt, VC) and the **rate of approach** (which is determined by the turnover rate $\frac{R}{V}$).

   Note: Two of the differential equations provided do not match any of the graphs. If an equation has no match, write "None".

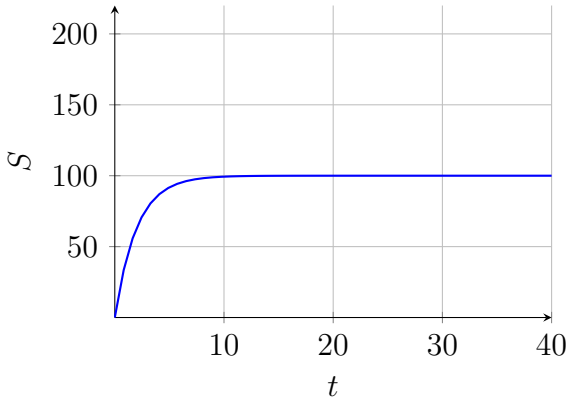
(a) $\frac{dS}{dt} = 50 - 0.5S$ (c) $\frac{dS}{dt} = 10 - 0.5S$ (e) $\frac{dS}{dt} = 100 - 0.5S$

(b) $\frac{dS}{dt} = 10 - 0.1S$ (d) $\frac{dS}{dt} = 20 - 0.1S$ (f) $\frac{dS}{dt} = 50 - 0.1S$

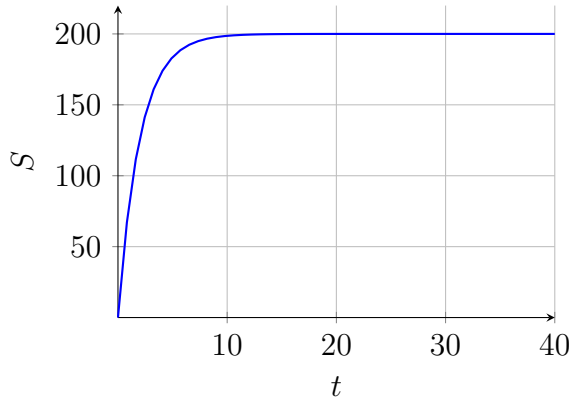
i)



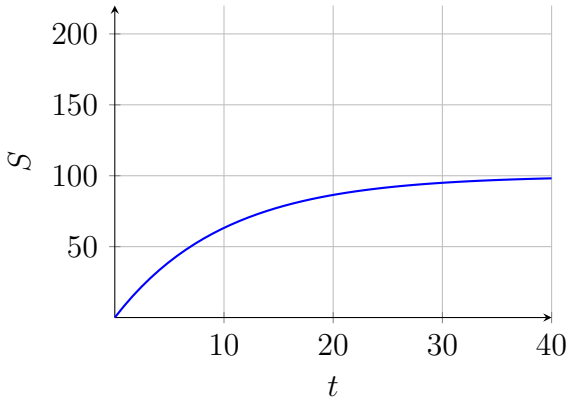
ii)



iii)



iv)



6. Match each physical mixing scenario described below with the differential equation that models the amount of salt $S(t)$ in the tank.

   Assume the flow rate in (R) equals the flow rate out (R), so the total volume (V) remains constant. Let C represent the incoming concentration.

   Note: Two of the differential equations provided do not match any of the scenarios. If an equation has no match, write "None".

   (a) $\frac{dS}{dt} = 50 - 0.01S$ (c) $\frac{dS}{dt} = 50 - 0.05S$ (e) $\frac{dS}{dt} = 10 - 0.05S$

   (b) $\frac{dS}{dt} = 5 - 0.01S$ (d) $\frac{dS}{dt} = -0.05S$ (f) $\frac{dS}{dt} = 5 - 0.1S$

   i) A 1000 L tank. Brine containing 5 kg of salt per liter is pumped in at a rate of 10 L/min.

   ii) A 100 L tank. Brine containing 5 kg of salt per liter is pumped in at a rate of 1 L/min.

   iii) A 1000 L tank. Brine containing 1 kg of salt per liter is pumped in at a rate of 50 L/min.

   iv) A 100 L tank full of brine. **Pure water** is pumped in at a rate of 5 L/min.

7. A 500-gallon tank initially contains 400 gallons of water with 50 lbs of salt dissolved in it. Brine containing 0.2 lbs of salt per gallon is pumped into the tank at a rate of 5 gallons per minute. The well-mixed solution is pumped out *at a faster rate* of 7 gallons per minute. Let $S(t)$ be the amount of salt in the tank at time t in minutes.

(a) Write down (but do not solve) an initial value problem for $S(t)$.

(b) What is the domain of this problem? (In other words, how long can this situation continue?)

(c) Does this differential equation have any equilibria? Show your work.

8. A 1000-liter vat initially contains 200 liters of pure water. A liquid dye solution containing 3 kg of dye per liter is pumped into the vat at a rate of 10 liters per minute. The well-mixed solution *is drained out at a slower rate* of 6 liters per minute. Let $S(t)$ be the amount of dye in the vat at time t in minutes.

(a) Write down (but do not solve) an initial value problem for $S(t)$.

(b) What is the domain of this problem? (In other words, how long until the vat overflows?)

(c) Does this differential equation have any equilibria? Show your work.

Chemical Rates

1. Consider the following model of a chemical reaction. First, make conjectures regarding how it proceeds, then solve.

$$\begin{aligned} \frac{d[A]}{dt} &= -[A] + 4[B], \quad [A](0) = 5 \\ \frac{d[B]}{dt} &= [A] - 4[B], \quad [B](0) = 0 \end{aligned}$$

   What happens to the solutions as $t \to \infty$?

2. Consider the following model of a chemical reaction. First, make conjectures regarding how it proceeds, then solve.

$$\begin{aligned} \frac{d[A]}{dt} &= -4[A] + 2[B], \quad [A](0) = 6 \\ \frac{d[B]}{dt} &= 4[A] - 2[B], \quad [B](0) = 0 \end{aligned}$$

   What happens to the solutions as $t \to \infty$?

3. Consider the following model of a chemical reaction. First, make conjectures regarding how it proceeds, then solve.

$$\begin{aligned} \frac{d[A]}{dt} &= -2[A] + 5[B], \quad [A](0) = 0 \\ \frac{d[B]}{dt} &= 2[A] - 5[B], \quad [B](0) = 14 \end{aligned}$$

   What happens to the solutions as $t \to \infty$?

4. Suppose that we have $k_1 = 1$ and $k_2 = 2$, but that the initial concentrations, A_0 and B_0, are not known. Make conjectures as to what the final concentrations of A and B will be, in terms of A_0 and B_0. Now solve the equations above. Your solution will depend on A_0 and B_0. What happens to the solutions as $t \to \infty$?

5. Suppose that we have $k_1 = 5$ and $k_2 = 3$, but that the initial concentrations, A_0 and B_0, are not known. Make conjectures as to what the final concentrations of A and B will be, in terms of A_0 and B_0. Now solve the equations above. Your solution will depend on A_0 and B_0. What happens to the solutions as $t \to \infty$?

Converting Descriptions to DEs

For each of the following scenarios, translate the verbal description into a differential equation. Be sure to define your variables and use k for any unspecified positive constants of proportionality.

1. A student is memorizing a vocabulary list of 500 words. The rate at which the student learns new words is proportional to the number of words they have not yet learned. Write a differential equation for $W(t)$, the number of words learned at time t.

2. A skydiver falls through the air. The rate of change of their downward velocity is driven by the constant acceleration of gravity, g, but it is reduced by air resistance. The air resistance is proportional to the square of their current velocity. Write a differential equation for the downward velocity $v(t)$.

3. An IV drip delivers a painkiller into a patient's bloodstream at a constant rate of 15 mg per hour. Simultaneously, the patient's liver clears the painkiller from the bloodstream at a rate proportional to the current amount of the drug in the blood. Write a differential equation for $M(t)$, the amount of painkiller in the bloodstream at time t.

4. In a town with a population of 25,000, a new rumor is spreading. The rate at which the rumor spreads is proportional to the product of the number of people who have heard it and the number of people who have *not* yet heard it. Write a differential equation for $P(t)$, the number of people who have heard the rumor.

5. A retirement account grows with a continuously compounded interest rate of 6% per year. At the same time, the retiree continuously withdraws living expenses at a constant rate of \$45,000 per year. Write a differential equation for $B(t)$, the account balance in dollars at time t in years.

6. A spherical raindrop evaporates as it falls. Its volume decreases at a rate proportional to its exposed surface area. Given that the surface area of a sphere is proportional to its volume raised to the two-thirds power ($A \propto V^{2/3}$), write a differential equation for the volume $V(t)$ of the raindrop.

7. The temperature of a hot cup of coffee decreases at a rate proportional to the difference between its own temperature and the constant room temperature of 72°F. Write a differential equation for the temperature of the coffee, $T(t)$.

8. In a certain chemical reaction, reactant A turns into product X. The reaction is "second-order," meaning the rate at which the product X forms is proportional to the square of the amount of reactant A currently remaining. If the initial amount of reactant A was A_0, and each unit of A perfectly converts into one unit of X, write a differential equation for the amount of product $X(t)$.